

SATQUERY AI

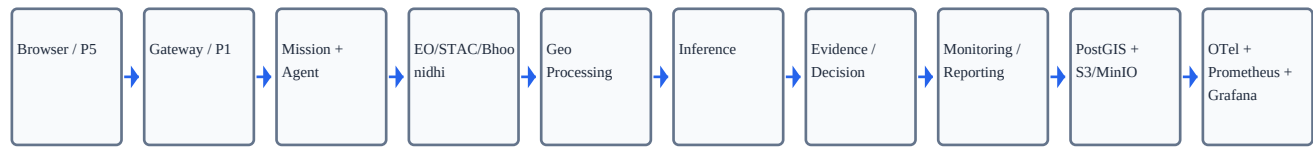
MASTER — COMPLETE ENGINEERING + INTEGRATION PRD

Six-person parallel execution system • Deadline 20 September 2026

Source of truth for issues, dependencies, integration, testing, security, observability, CI/CD, release and the five-minute flagship demo.

1. Executive Architecture

Overall architecture



The architecture preserves the supplied ~8-service boundary: Gateway/BFF, Mission, Agent, EO Data, Geo Processing, Inference, Evidence/Decision, Monitoring/Reporting. Redis/Celery/Streams provides prototype job/event plumbing. No service reads another service's private tables.

2. Six-Person Ownership Matrix

Person	Role	Primary ownership	Backup/review
P1	Tech Lead + Backend Architect	Gateway/BFF, Auth, org/RBAC, contracts, Mission Service	Mission backup; cross-service architecture
P2	AI / Agent Engineer	LangGraph, Mission Copilot, intent, tools, sensor arbitration, Evidence/Confidence	Evidence; AI regression
P3	ML / Computer Vision Engineer	Optical/SAR/temporal models, inference, fusion, evaluation	Geo processing; model evaluation
P4	Geospatial / Data Engineer	Bhoonidhi/STAC, normalization, COG, GDAL/Rasterio, PostGIS, TiTiler	Monitoring data pipeline
P5	Frontend / Geospatial UX Engineer	Next.js, MapLibre, mission console, evidence/timeline/report UX	API integration
P6	Platform / DevOps + QA Engineer	Docker, CI/CD, observability, test automation, security, release	Monitoring/alerts; release

3. Who Talks to Whom

Relationship	Reason	Artifact
P1↔P2	Mission/run lifecycle, agent plan/execute, tools, evidence	Mission + Agent API/events
P1↔P3	Inference job/model/provenance	InferenceRequest/Result
P1↔P4	Observation/AssetRef/GeoJob	OpenAPI + fixtures
P1↔P5	Gateway DTOs/generated client	OpenAPI/generated TS
P1↔P6	CI, release, health, security	Pipeline/runbook
P2↔P3	Sensor/model fit, evidence confidence	SensorDecision + InferenceResult
P2↔P4	Search, quality, acquisition	ObservationSearch + AssetRef
P3↔P4	CRS/resolution/preprocessing	GeoJobResult + AssetRef
P4↔P5	Tiles, GeoJSON, masks, AOI	Tile/Geo contracts
P5↔P6	E2E/demo/security	Playwright + DEMO

4. Contract Freeze Matrix

Contract	Producer	Consumers	Freeze
Mission OpenAPI	P1	P2/P5/P6	Sep 11
Observation + AssetRef	P4	P2/P3/P5	Sep 11
GeoJob	P4	P3/P5/P6	Sep 11
Inference	P3	P2/P5/Evidence	Sep 11
Evidence	P2/P3	P5/P1/P6	Sep 12
Monitoring/Reporting	P6/P4	P1/P5/P2	Sep 13
ErrorResponse	P1	ALL	Sep 11
EventEnvelope	P1	ALL	Sep 11

5. Integration Invariants

- No inference without authorized AssetRef + mission_id.
- No evidence without provenance.

- Low confidence never becomes confident prose.
- No cross-service private DB reads.
- organization_id is server-derived.
- Async jobs are idempotent/deduplicated.
- Events are versioned and traced.
- Browser calls Gateway only.

6. Complete Master Issue Register

Total implementation issues: 108. Every issue from P1–P6 is reproduced below; this is the master backlog.

Issue	Title	Owner	Service	Pri	Deps	Contact
P1-01	Monorepo and service skeletons	P1	Gateway / Mission	P0	None	P6
P1-02	Freeze canonical API, error and event contracts	P1	Gateway / Mission	P0	P1-01	P2
P1-03	OAuth2/OIDC JWT verification foundation	P1	Gateway / Mission	P0	P1-01	P5
P1-04	RBAC and tenant isolation	P1	Gateway / Mission	P0	P1-03	P6
P1-05	API Gateway/BFF routing, CORS and rate limits	P1	Gateway / Mission	P0	P1-04	P2
P1-06	Mission and AOI lifecycle service	P1	Gateway / Mission	P0	P1-05	P5
P1-07	Mission run/job lifecycle bridge	P1	Gateway / Mission	P0	P1-06,P6-02	P6
P1-08	Mission WebSocket status stream	P1	Gateway / Mission	P1	P1-07,P6-02	P2
P1-09	Audit trail and trace metadata	P1	Gateway / Mission	P0	P1-08,P6-02	P5
P1-10	Shared internal client and error mapping	P1	Gateway / Mission	P0	P1-09,P6-02	P6
P1-11	Service-to-service identities and scopes	P1	Gateway / Mission	P0	P1-10	P2
P1-12	Idempotency and duplicate execution controls	P1	Gateway / Mission	P0	P1-11	P5
P1-13	Required architecture ADR set	P1	Gateway / Mission	P1	P1-12	P6
P1-14	OpenAPI/event compatibility CI gate	P1	Gateway / Mission	P0	P1-13	P2
P1-15	Mission ↔ Agent integration	P1	Gateway / Mission	P0	P1-14,P2-05	P5
P1-16	Generated frontend client verification	P1	Gateway / Mission	P0	P1-15,P2-05	P6
P1-17	Timeout/retry/circuit-breaker policy	P1	Gateway / Mission	P1	P1-16,P2-05	P2
P1-18	Final architecture hardening and sign-off	P1	Gateway / Mission	P0	P1-17,P2-05	P5
P2-01	Agent service skeleton and MissionState	P2	Agent	P0	P1-01	P1
P2-02	Input sanitizer and intent validator	P2	Agent	P0	P1-01,P1-02	P4
P2-03	Intent extraction and mission planning	P2	Agent	P0	P1-01,P1-02	P3
P2-04	Async agent execute endpoint and run orchestration	P2	Agent	P0	P2-03	P1
P2-05	Versioned tool registry and schemas	P2	Agent	P0	P2-04	P4
P2-06	Permission-aware tool executor and budgets	P2	Agent	P0	P2-05	P3
P2-07	Observation search and asset selection tools	P2	Agent	P0	P2-06,P1-02	P1
P2-08	Temporal planning and previous/current selection	P2	Agent	P0	P2-07,P1-02	P4
P2-09	Adaptive sensor arbitration	P2	Agent	P0	P2-08,P1-02	P3
P2-10	Evidence object / graph builder	P2	Agent	P0	P2-09,P1-02	P1
P2-11	Confidence and uncertainty gate	P2	Agent	P0	P2-10	P4
P2-12	Sensor disagreement analysis	P2	Agent	P1	P2-11	P3
P2-13	Autonomous evidence acquisition loop	P2	Agent	P1	P2-12	P1
P2-14	Evidence-only output synthesizer and WHY explanation	P2	Agent	P0	P2-13	P4
P2-15	Prompt/evaluation harness and agent regression	P2	Agent	P0	P2-14,P4-05	P3
P2-16	Agent observability and tool-call audit	P2	Agent	P0	P2-15,P4-05	P1
P2-17	Agent failure/recovery policies	P2	Agent	P1	P2-16,P4-05	P4
P2-18	Agent release hardening and pinned demo profile	P2	Agent	P0	P2-17,P4-05	P3
P3-01	Inference service skeleton and model registry	P3	Inference	P0	P1-01	P4
P3-02	Asset preflight and input validation	P3	Inference	P0	P1-01,P1-02	P2
P3-03	Optical inference adapter	P3	Inference	P0	P1-01,P1-02	P6
P3-04	SAR inference adapter	P3	Inference	P0	P3-03	P4
P3-05	Temporal semantic change-detection pipeline	P3	Inference	P0	P3-04	P2

Issue	Title	Owner	Service	Pri	Deps	Contact
P3-06	Cross-modal optical-SAR fusion	P3	Inference	P0	P3-05	P6
P3-07	Inference result postprocessor	P3	Inference	P0	P3-06,P4-02	P4
P3-08	Model artifact verification and version pinning	P3	Inference	P0	P3-07,P4-02	P2
P3-09	Model serving optimization / quantization profile	P3	Inference	P1	P3-08,P4-02	P6
P3-10	Async inference worker and progress	P3	Inference	P0	P3-09,P4-02	P4
P3-11	Confidence calibration and model metrics	P3	Inference	P1	P3-10	P2
P3-12	Infrastructure intersection output adapter	P3	Inference	P1	P3-11	P6
P3-13	Flood composite analysis pipeline	P3	Inference	P0	P3-12	P4
P3-14	Inference failure, retry and degraded mode	P3	Inference	P0	P3-13	P2
P3-15	ML evaluation suite	P3	Inference	P0	P3-14,P2-05	P6
P3-16	Inference observability and trace propagation	P3	Inference	P0	P3-15,P2-05	P4
P3-17	Model/data provenance manifest	P3	Inference	P0	P3-16,P2-05	P2
P3-18	Pinned inference release profile	P3	Inference	P0	P3-17,P2-05	P6
P4-01	EO/Data service skeleton and provider adapter interface	P4	EO Data / Geo	P0	P1-01	P1
P4-02	Provider configuration and secret boundary	P4	EO Data / Geo	P0	P1-01,P1-02	P3
P4-03	Observation and AssetRef domain model	P4	EO Data / Geo	P0	P1-01,P1-02	P2
P4-04	STAC adapter	P4	EO Data / Geo	P0	P4-03	P1
P4-05	Bhoonidhi adapter	P4	EO Data / Geo	P0	P4-04	P3
P4-06	Observation normalization pipeline	P4	EO Data / Geo	P0	P4-05	P2
P4-07	Spatial/temporal observation search	P4	EO Data / Geo	P0	P4-06,P1-02	P1
P4-08	Asset resolver	P4	EO Data / Geo	P0	P4-07,P1-02	P3
P4-09	Secure asset retrieval and content validation	P4	EO Data / Geo	P0	P4-08,P1-02	P2
P4-10	Raster validation	P4	EO Data / Geo	P0	P4-09,P1-02	P1
P4-11	CRS normalization and reprojection	P4	EO Data / Geo	P0	P4-10	P3
P4-12	AOI clipping and windowed processing	P4	EO Data / Geo	P0	P4-11	P2
P4-13	COG generation and overviews	P4	EO Data / Geo	P0	P4-12	P1
P4-14	PostGIS spatial operations	P4	EO Data / Geo	P0	P4-13	P3
P4-15	TiTiler integration	P4	EO Data / Geo	P1	P4-14,P3-05	P2
P4-16	Async GeoJob worker	P4	EO Data / Geo	P0	P4-15,P3-05	P1
P4-17	Geo failure recovery and fixture fallback	P4	EO Data / Geo	P0	P4-16,P3-05	P3
P4-18	Monitoring observation selection support	P4	EO Data / Geo	P1	P4-17,P3-05	P2
P4-19	EO data quality scoring/enrichment	P4	EO Data / Geo	P1	P4-18,P3-05	P1
P4-20	Pinned EO/Geo fixture pack and release hardening	P4	EO Data / Geo	P0	P4-19,P3-05	P3
P5-01	Next.js application shell and design system	P5	Web / Map UX	P0	P1-01	P1
P5-02	Generated API client integration	P5	Web / Map UX	P0	P1-01,P1-02	P4
P5-03	Mission console query and planning UX	P5	Web / Map UX	P0	P1-01,P1-02	P2
P5-04	Mission status and run timeline	P5	Web / Map UX	P0	P5-03	P1
P5-05	MapLibre map workspace	P5	Web / Map UX	P0	P5-04	P4
P5-06	AOI draw/edit and validation UX	P5	Web / Map UX	P0	P5-05	P2
P5-07	Observation and before/after comparison	P5	Web / Map UX	P0	P5-06,P1-02	P1
P5-08	Change polygons, masks and confidence visualization	P5	Web / Map UX	P0	P5-07,P1-02	P4
P5-09	WHY? evidence panel	P5	Web / Map UX	P0	P5-08,P1-02	P2
P5-10	Sensor arbitration and disagreement UI	P5	Web / Map UX	P1	P5-09,P1-02	P1
P5-11	Confidence/uncertainty and human-in-loop UX	P5	Web / Map UX	P0	P5-10	P4
P5-12	Mission memory and temporal history	P5	Web / Map UX	P0	P5-11	P2
P5-13	Persistent monitoring mission screen	P5	Web / Map UX	P0	P5-12	P1
P5-14	Report generation and report view	P5	Web / Map UX	P0	P5-13	P4
P5-15	Admin, security and accessibility UX	P5	Web / Map UX	P1	P5-14,P4-05	P2
P5-16	Frontend performance and error boundaries	P5	Web / Map UX	P1	P5-15,P4-05	P1
P5-17	E2E UX hardening and deterministic demo mode	P5	Web / Map UX	P0	P5-16,P4-05	P4

Issue	Title	Owner	Service	Pri	Deps	Contact
P6-01	Docker Compose local integration environment	P6	Platform / QA / SRE	P0	P1-01	P1
P6-02	CI lint/type/unit pipeline	P6	Platform / QA / SRE	P0	P1-01,P1-02	P5
P6-03	Security scanning baseline	P6	Platform / QA / SRE	P0	P1-01,P1-02	P3
P6-04	Contract test harness and generated fixtures	P6	Platform / QA / SRE	P0	P6-03	P1
P6-05	DEV deployment and service health smoke	P6	Platform / QA / SRE	P0	P6-04	P5
P6-06	OpenTelemetry collector and structured logging	P6	Platform / QA / SRE	P0	P6-05	P3
P6-07	Prometheus metrics and Grafana dashboards	P6	Platform / QA / SRE	P0	P6-06,P1-02	P1
P6-08	Cross-service integration test suite	P6	Platform / QA / SRE	P0	P6-07,P1-02	P5
P6-09	Flagship flood E2E automation	P6	Platform / QA / SRE	P0	P6-08,P1-02	P3
P6-10	Performance/load test harness	P6	Platform / QA / SRE	P1	P6-09,P1-02	P1
P6-11	Security/tenant/prompt-injection regression suite	P6	Platform / QA / SRE	P0	P6-10	P5
P6-12	Failure injection and recovery tests	P6	Platform / QA / SRE	P0	P6-11	P3
P6-13	Secrets/config and least-privilege hardening	P6	Platform / QA / SRE	P0	P6-12	P1
P6-14	Monitoring/report integration QA	P6	Platform / QA / SRE	P0	P6-13	P5
P6-15	Release candidate, immutable images and rollback	P6	Platform / QA / SRE	P0	P6-14,P5-05	P3
P6-16	Two full demo rehearsals and operator runbook	P6	Platform / QA / SRE	P0	P6-15,P5-05	P1
P6-17	Final QA sign-off and release gate	P6	Platform / QA / SRE	P0	P6-16,P5-05	P5

7. Dependency Graph and Critical Path

Flagship critical path



Critical path: P1 contract freeze → P4 observations/AssetRef → P4 Geo processing → P3 inference → P2 evidence/confidence → P5 map/evidence → P6 E2E/release. Everything else is parallelized around this path.

8. Parallel Workstreams

Track	Owner	Parallel work	Converges at
Platform	P6	Compose, CI, security, OTel, mocks	DEV + release gates
Backend	P1	Gateway, Auth, Mission, contracts	Mission↔Agent
Agent	P2	Graph, intent, tools, sensor, evidence	Data/Inference/Evidence
ML	P3	Model adapters, temporal, fusion, eval	InferenceResult
Geo/Data	P4	STAC/Bhoonidhi, COG, PostGIS, tiles	AssetRef/GeoJob
Frontend	P5	Shell, client, map, evidence UX	Gateway/Evidence

9. Contract-First Development

- Create contract → review → freeze → generate mock/client → producer implementation → consumer implementation → contract CI → integration test → E2E.
- Breaking changes are never silent: version API/event schema, update fixtures, migrate consumers, then merge.
- When an upstream issue is incomplete, use the contract fixture and continue; no developer should sit idle.

10. Mock / Stub / Fixture Strategy

Dependency	Fixture/mock	Failures simulated	Owner
EO Data	Pinned STAC/Bhoonidhi JSON + Observation fixtures	401/403/429/5xx/timeout	P4/P6
Geo	Small GeoTIFF/COG + GeoJob stub	bad CRS/AOI/disk/timeout	P4/P6
Inference	InferenceResult fixtures	timeout/OOM/model unavailable	P3/P6
Agent	Node/tool fixtures	denied tool/injection/schema mismatch	P2/P6
Gateway/Mission	OpenAPI mock server	401/403/409/429/5xx	P1/P6
Frontend	Generated client + mocked responses	loading/error/partial state	P5/P6
Monitoring/Report	Job/report fixtures	scheduler/report failure	P6

11. Shared Issue Workflow

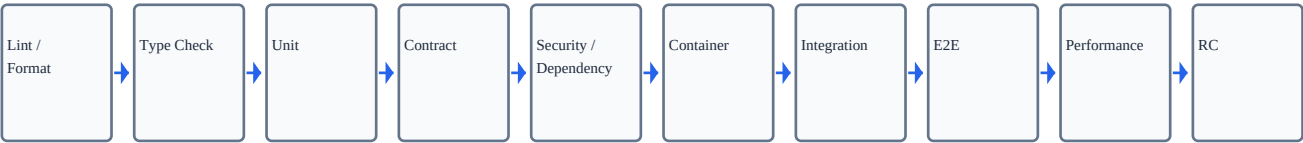
Kanban workflow



BLOCKED is a side lane. A blocked issue must name the blocker owner, the contract/fixture that can unblock it and the next escalation. Escalate after 30–60 minutes.

12. CI/CD

PR → release



P6 owns pipeline mechanics; domain owners own failing tests; P1 approves architecture/contract exceptions. Merge to develop deploys DEV; RC is produced after feature freeze and critical gates.

13. Testing Pyramid

Layer	Purpose	Contributors	Gate
Unit	Business logic and validators	All	PR
Contract	API/event compatibility	P1/P6 + producers	PR
Integration	Real service boundaries	All + P6	DEV
Service E2E	Domain workflows	P2/P3/P4	DEV
Product E2E	Full flood mission	P5/P6 + all	RC/release
Load	Latency/throughput	P6/P1/P3/P4	RC
Security	Auth/RBAC/injection/SSRF/files	P1/P2/P4/P6	PR/RC
Failure	Provider/model/Redis/partial pipeline	P6 + domain owners	RC

14. Ten End-to-End Scenarios

#	Starting/action	API/event chain	Expected result
1	Natural language → flood mission	Submit Guntur flood query	Mission→Agent→Data→Geo→Inference→Evidence
2	Construction change	Ask what changed	Mission→Data→Temporal→Evidence
3	Optical/SAR disagreement	Use disagreement fixture	Sensor→Inference→Evidence→Gate
4	Low confidence	Confidence 0.47 fixture	Evidence→Gate
5	Bhoonidhi unavailable	Provider failure profile	Agent→P4→fixture
6	Inference failure	Timeout/OOM fixture	Geo→P3→recovery
7	Invalid AOI	Malformed/self-intersecting GeoJSON	Gateway→Mission/P4
8	Unauthorized tenant	User B accesses User A mission	Gateway→RBAC
9	Monitoring alert	Threshold crossing fixture	Monitoring→Data→Analysis→Alert
10	Report generation	Generate from completed mission	Report→Evidence/Decision

15. Security Architecture

Defense in depth



Control	Owner	Implementation
OAuth2/OIDC/JWT	P1	Issuer/aud/signature/expiry/algorithm validation
RBAC/tenant	P1/P6	Server-derived org context + negative tests
Prompt injection	P2/P6	Sanitization + task/tool allowlist + regression
SSRF/files/raster	P4/P6	Domain allowlist, magic bytes, size/path controls
Tool permissions	P2/P1	Scoped registry and budgets
Rate limits	P1/P6	Redis token bucket + payload limits
Secrets/TLS	P1/P6	Externalized secrets; no secret logs
Audit	P1/P6	Actor/tenant/trace/action; redaction
Containers/deps	P6	Least privilege + SAST/dependency/image scans

16. Observability

Every mission trace: user → mission → run → agent → tool calls → observation → geo job → inference → evidence → decision → report. Required fields: trace_id, mission_id, run_id, job_id, organization_id, model_version, dataset_id, schema version. OpenTelemetry provides traces; Prometheus metrics; Grafana dashboards.

17. Performance Targets

Target	Goal	Benchmark
Normal UI	<500 ms	Browser timing excluding heavy map/network loads
Metadata search	<2 s normal/cached	Pinned STAC + DEV p50/p95
Small analysis	<10 s where feasible	Pinned small raster/model profile
Large analysis	Async	202 + job status + completion
Queue latency	Bounded	enqueue→worker start p50/p95
Report	Async/idempotent	Duplicate request + artifact timing
Flagship E2E	Five-minute operator flow	Nightly deterministic run + two rehearsals

18. Release Plan — Sep 11–20

Date	Focus	Gate
Sep 11	Contract freeze; skeletons; fixtures; Compose	Contract CI green
Sep 12	APIs + mocks; UI shell; stubs	All services boot
Sep 13	Real service integration	Mission→Agent→Data
Sep 14	First vertical slice	Flood path works
Sep 15	Sensor/evidence/change	Differentiators visible
Sep 16	Security + failures	Negative/failure suite
Sep 17	Full regression	Nightly E2E green
Sep 18	FEATURE FREEZE + RC	No new features
Sep 19	Security/load/failure + 2 rehearsals	Release ready
Sep 20	FINAL RELEASE	Tagged release

19. Critical Scope

Tier	Protect
MUST HAVE	Natural language mission; Bhoonidhi/STAC/fixtures; optical+SAR+temporal; evidence/confidence; map; monitoring; report; trace/security
SHOULD HAVE	Decision prioritization refinements; richer alerts; NDVI/NDWI/object detection if stable
NICE TO HAVE	Additional analytics and noncritical polish
DEFER	Marketplace, payments, mobile, large-scale model training, 3D

20. Five-Minute Demo Pipeline

Time	Action	Owner	Evidence/innovation
0:00–0:30	Open Guntur mission console	P5/P2	Command center
0:30–1:00	Enter flood query + plan	P2/P5	Natural language → mission
1:00–1:30	Show AOI + data discovery	P4	India-first EO
1:30–2:00	Show cloud degradation + sensor choice	P2/P3	Adaptive sensor intelligence
2:00–3:00	Before/after + change + area	P3/P4/P5	Optical + SAR + time
3:00–3:30	WHY? evidence graph	P2/P5	Evidence-first AI
3:30–4:00	Affected roads/structures	P3/P4	Decision intelligence
4:00–4:30	Create Flood Watch	P6/P4/P5	Persistent mission
4:30–5:00	Generate report	P6/P5	Operational output

21. Release Checklist

- ☐ All services healthy in DEMO.
- ☐ Contracts compatible; migrations applied.
- ☐ Secrets externalized; security scan clean.
- ☐ E2E green; monitoring/tracing working.
- ☐ Fallback fixtures verified.
- ☐ Report generation verified.
- ☐ Tenant isolation and prompt injection tests passed.
- ☐ Model/data fixture versions pinned.
- ☐ Immutable image digests recorded.
- ☐ Rollback tested.
- ☐ Two rehearsals completed.
- ☐ Final tag created.

22. Final Consistency Audit

- Every issue has owner, reviewer, dependencies, contact, tests, acceptance and integration checkpoint.
- Every consumer has a producer contract.

- P3 inputs match P4 AssetRef; P4 outputs match P3/P5 expectations.
- P5 consumes only Gateway/BFF APIs.
- P6 can build/test all services.
- No unbounded synchronous cycle; event loops are bounded and traced.
- Mocks/fixtures prevent idle time.
- No critical feature is first introduced after RC.
- Security and observability have named owners.
- Flagship E2E is deterministic with live-provider fallback.

23. Operating Principles

WINNING OPERATING LOOP: PLAN → BUILD → INTEGRATE → DEMO → MEASURE → RETRO → REPLAN. WINNING ENGINEERING LOOP: CONTRACT → IMPLEMENT → TEST → OBSERVE → INTEGRATE → RELEASE. The agent is a coordinator and synthesizer; deterministic geospatial processing and specialist models produce structured evidence.

Appendix — Complete Issue-to-Owner Map

Owner	Issue range	Count
P1	P1-01 ... P1-18	18
P2	P2-01 ... P2-18	18
P3	P3-01 ... P3-18	18
P4	P4-01 ... P4-20	20
P5	P5-01 ... P5-17	17
P6	P6-01 ... P6-17	17